

Financial Goal Planner: An AI-Driven Goal-Based Investment Planning and Risk-Aware Recommendation System

Amarthya Regimon Joseph Aswin P Athira Raj Haleema S

Department of Computer Science and Engineering
CAPE College of Engineering Alappuzha

Under the guidance of:

Mr. Siju Koshy
Department of Computer Science and Engineering

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Introduction

This project proposes a **mobile application** that recommends personalized savings plans and investment allocations across mutual funds, equity, and debt instruments based on a user's financial goal, horizon, and risk profile

The system aims to provide adaptive, risk-aware, and horizon-sensitive investment guidance through a mobile-first interface.

Problem Statement

Individuals planning for financial goals – such as retirement, a child's education, or purchasing a vehicle – face difficulty determining **how much to save, for how long, and where to invest**, because the appropriate level of investment risk depends heavily on the goal's remaining time horizon.

Objectives

- ① To design a mobile application that allows users to define financial goals with a target amount and time horizon.
- ② To build a recommendation engine that dynamically adjusts asset allocation across mutual funds, equity, and debt instruments.
- ③ To provide an intuitive mobile dashboard for tracking goal progress, required SIP contributions, and recommended portfolio adjustments over time.

Literature Summary (1/4)

Ref	Author & Year	Title	Journal	Methodology	Result	Advantages	Disadvantages
1	Karzanov et al., 2025	Regret-Optimized Portfolio Enhancement [BASE PAPER]	arXiv	Regret-based portfolio optimization using risk-adjusted objectives and historical market data.	Produces portfolios with improved regret performance and better control of downside risk.	Improves portfolio robustness and risk control.	Requires reliable market estimates and can be computationally demanding.
2	Dhashanamoorthis et al., 2024	AI-driven robo-advisors on personal financial management	Int. J. of Sci. and Research	AI-based robo-advisory framework using financial analysis, investor profiling and personalized recommendations.	Demonstrates the potential of AI to automate personalized financial-management decisions.	Automated and personalized advice with reduced advisory effort.	Dependent on data quality and has limited human interaction.
3	Sunilkumar, 2025	AI Revolution in Wealth Management	J. of Info Systems Eng.	Analysis of AI and machine-learning applications in wealth-management and investment decisions.	Shows that AI can improve automation, financial analysis and decision-support capabilities.	Enables automation, scalability and faster financial analysis.	Data dependency, model complexity and explainability concerns.
4	Ansari et al., 2024	Multifaceted Approach to Stock Market Trading Using RL	IEEE Access	Reinforcement-learning framework using market information and trading strategies for automated stock trading.	RL-based strategies demonstrate the ability to adapt trading decisions to changing market conditions.	Adaptive decision making and automated trading.	Sensitive to market noise, reward design and training parameters.
5	Sattar et al., 2025	RMS-Driven Deep RL for Optimized Portfolio Management	IEEE Access	Deep reinforcement learning combined with risk-management techniques for dynamic portfolio optimization.	Demonstrates improved portfolio optimization by incorporating risk management into the learning process.	Combines return optimization with risk management.	High computational requirements and possible training instability.
6	Yin & Guo, 2025	Complex forecasting via chain-of-thought of LLMs	Science Direct	Large language models with chain-of-thought reasoning for complex forecasting tasks.	Shows that structured LLM reasoning can improve performance on complex forecasting problems.	Improves reasoning capability for complex forecasting.	High computational cost and difficulty verifying generated reasoning.

Literature Summary (2/4)

Ref	Author & Year	Title	Journal	Methodology	Result	Advantages	Disadvantages
7	Benhamou et al., 2020	Detecting and adapting to crisis pattern with context DRL	arXiv	Context-aware deep reinforcement learning for detecting and adapting to financial crisis patterns.	The framework enables portfolio strategies to adapt when crisis-related market patterns are detected.	Adapts portfolio decisions during changing and crisis conditions.	Requires sufficient crisis-period data and complex training.
8	Samanta et al., 2024	Effectiveness of SIP Investment Among Retail Investors	Int. J. for Novel Research	Empirical analysis of systematic investment plans using investor data and statistical analysis.	The study indicates the role of systematic investing in promoting disciplined investment behaviour among retail investors.	Highlights disciplined investing and investor behaviour.	Limited applicability to automated portfolio optimization.
9	Kanagaraj et al., 2024	Potential of AI in personal financial planning	arXiv	AI-based personal financial planning using financial information and intelligent recommendation techniques.	Shows that AI can support personalized financial planning and automated recommendations.	Supports personalized planning and automated decision making.	Privacy, data-quality and explainability concerns.
10	Boaretto & Medeiros, 2023	Forecasting inflation using disaggregates and ML	arXiv	Machine-learning models using disaggregated economic variables for inflation forecasting.	Machine-learning approaches demonstrate the ability to capture nonlinear relationships in inflation forecasting.	Captures complex economic relationships.	Sensitive to data quality, feature selection and model choice.
11	Solanki & Jha, 2025	GAF-ResNet-LSTM Framework for Asset Selection	IEEE Access	Gramian Angular Field representation combined with ResNet and LSTM for financial asset selection.	The hybrid framework effectively captures spatial and temporal patterns for asset-selection decisions.	Captures both spatial and temporal financial patterns.	High model complexity and computational requirements.
12	Kevin & Yugo-puspito, 2025	Hybrid LSTM and PPO Networks for Portfolio Optimization	arXiv	Hybrid architecture combining LSTM prediction with PPO reinforcement learning for portfolio optimization.	The hybrid model combines sequential market prediction with adaptive portfolio allocation.	Combines prediction with adaptive portfolio decisions.	Training complexity and sensitivity to hyperparameters.

Literature Summary (3/4)

Ref	Author & Year	Title	Journal	Methodology	Result	Advantages	Disadvantages
13	Olamijuwon & Zouo, 2024	ML in budget forecasting for corporate finance	Open Access Research	Machine-learning models applied to corporate budget and financial forecasting.	ML-based forecasting can improve automated budget prediction and identify complex financial patterns.	Automates forecasting and identifies complex patterns.	Dependent on historical data quality and model assumptions.
14	Spadea & Seneviratne, 2025	Parallel and Multi-Stage Knowledge Graph Retrieval	arXiv	Parallel and multi-stage knowledge-graph retrieval for structured information processing.	The approach improves information retrieval by combining multiple retrieval stages and contextual knowledge.	Improves retrieval efficiency and contextual information access.	Increased system complexity and computational overhead.
15	Patra et al., 2025	AI-Driven Goal Based Financial Planning System	Rademics Research Inst.	AI-based financial planning using user goals, financial information and automated recommendations.	The system supports goal-oriented financial planning through personalized AI-based recommendations.	Provides personalized and goal-oriented financial planning.	Requires detailed user information and reliable financial data.
16	Lwele et al., 2025	Risk-Aware Deep RL for Dynamic Portfolio Optimization	arXiv	Risk-aware deep reinforcement learning for dynamic portfolio allocation.	The framework incorporates portfolio risk into the learning process to support risk-sensitive allocation decisions.	Explicitly incorporates investment risk.	Complex training and sensitivity to market conditions.
17	Gaspar & Oliveira, 2024	Robo Advising and Investor Profiling	MDPI	Robo-advisory framework using investor profiling and automated portfolio recommendations.	Shows how investor profiles can be incorporated into automated robo-advisory recommendations.	Provides scalable and personalized investment advice.	May provide less personalization than human advisory.
18	Shetty et al., 2026	Robo-advisors in financial services	Science Direct	Review and analysis of AI-powered robo-advisory systems in financial services.	The review identifies automation, accessibility and scalability as major capabilities of robo-advisory systems.	Highlights automation, accessibility and scalability.	Trust, transparency, regulation and explainability concerns.

Literature Summary (4/4)

Ref	Author & Year	Title	Journal	Methodology	Result	Advantages	Disadvantages
19	Kofman, 2025	Scoring the Ethics of AI Robo-Advice	J. of Business Ethics	Ethical evaluation framework for assessing AI-based robo-advisory practices.	Provides a structured approach for identifying and evaluating ethical issues in AI-based robo-advice.	Provides a framework for ethical evaluation.	Ethical assessment may depend on subjective criteria.
20	Lamukanyani et al., 2025	Sentiment-Aware Portfolio Optimization (CVaR-Based)	IEEE Access	Portfolio optimization combining market sentiment with CVaR-based downside-risk management.	The framework incorporates investor sentiment and downside risk into portfolio allocation decisions.	Combines sentiment analysis with downside-risk control.	Dependent on sentiment quality and accurate CVaR estimation.
21	Ahn & Kang, 2025	Simul-RL Portfolio Framework: BSM and RL	IEEE Access	Simulation framework combining Black-Scholes-Merton modelling with reinforcement learning.	Demonstrates how financial modelling and RL can be integrated for portfolio decision making.	Combines financial modelling with adaptive RL decisions.	Simulation assumptions may not fully represent real markets.
22	Pereguda et al., 2024	Impact of the Digital Economy on Investment Portfolio Mgmt	Eastern Euro. Union	Analysis of digital-economy technologies and their impact on investment portfolio management.	Shows the influence of digital technologies on modern portfolio-management processes.	Highlights potential benefits of digitalization.	Primarily analytical with limited direct implementation.
23	Kulkarni et al., 2020	Role of robo-advisors in behavioural finance	Cogent OA	Analysis of robo-advisory systems from a behavioural-finance perspective.	Shows the relationship between automated investment services and behavioural aspects of investors.	Connects automated investing with investor behaviour.	Behavioural factors are difficult to quantify and model.
24	Enkhsaikhan & Jo, 2024	Uncertainty-Aware RL for Portfolio Optimization	IEEE Access	Reinforcement-learning framework incorporating uncertainty into portfolio optimization.	The framework accounts for uncertainty during decision making and allocation.	Accounts for uncertainty in portfolio decisions.	Higher computational complexity and difficult uncertainty estimation.

Literature Review (1/24): Regret-Optimized Portfolio Enhancement through DRL and Future Looking Rewards

2025 – Base Paper

- **Relation:** This is the base paper – our project directly adapts its regret-optimized, future-looking reward formulation for goal-based, horizon-sensitive asset allocation.
- **Methodology:** PPO-based DRL framework for dynamic portfolio rebalancing using a regret-based reward (relative to a 60/40 benchmark) combined with a future-looking (forward-horizon) reward component; evaluated via Sharpe ratio and maximum drawdown.

Literature Review (2/24): AI-driven robo-advisors on personal financial management and investment strategies

2024

- **Relation:** Establishes the general case for AI-driven personalized investment advice that our app builds on, but uses no RL-based reward design.
- **Methodology:** Conceptual/survey-style analysis of how robo-advisors use algorithms and ML on market and client data to create tailored, automated investment plans.

Literature Review (3/24): AI Revolution in Wealth Management: Applications and Horizons

2025

- **Relation:** Provides industry-level context on AI adoption in wealth management, motivating the case for an AI-driven planning app, though not a DRL-based technical contribution.
- **Methodology:** Industry-perspective review discussing AI applications (compliance automation, personalization, back-office automation) in wealth management.

Literature Review (4/24): A Multifaceted Approach to Stock Market Trading Using Reinforcement Learning

2024

- **Relation:** Shares the RL-for-finance foundation with the base paper, but focuses on multi-stock trading rather than goal-based portfolio allocation.
- **Methodology:** Uses Advantage Actor-Critic (A2C) and Deep Deterministic Policy Gradient (DDPG) with a novel risk-aware reward (PSR) combining technical and fundamental indicators; validated on 30 Dow Jones stocks.

Literature Review (5/24): A Novel RMS-Driven Deep RL for Optimized Portfolio Management in Stock Trading

2025

- **Relation:** Closely related to the base paper's risk-aware reward design philosophy – both incorporate drawdown-based rewards, though this paper adds sentiment rather than regret/future-looking terms.
- **Methodology:** Proposes PPO_RMS, A2C_RMS, and DDPG_RMS models combining earnings-report data, Max Drawdown-based rewards, and sentiment indicators (RMS); DDPG_RMS outperforms the DJI benchmark.

Literature Review (6/24): Complex forecasting and investment strategy optimization via chain-of-thought of LLMs

2025

- **Relation:** Explores an alternative (LLM-based) path to investment decision-making, contrasting with the base paper's RL-based approach; useful for comparing reasoning-driven vs. reward-driven strategies.
- **Methodology:** Applies chain-of-thought (CoT) prompting with ChatGPT-4.0 on anonymized financial statements of 608 Chinese A-share firms (2010–2023) to predict earnings direction and backtest a trading strategy.

Literature Review (7/24): Detecting and adapting to crisis pattern with context based DRL

2020

- **Relation:** An alternative robustness strategy to the base paper – uses contextual macro features instead of regret/future-looking reward shaping to handle adverse market regimes.
- **Methodology:** Two sub-network DRL architecture; one network processes past strategy performance/volatility, the other processes macro-context features (risk aversion, economic surprise index, cross-asset correlation), with adversarial training for robustness.

Literature Review (8/24): Effectiveness of SIP Investment Among Retail Investors in Jharkhand

2024

- **Relation:** Grounds our project's Indian retail-investor context and validates SIP as the primary savings vehicle our app should recommend for goal-based plans.
- **Methodology:** Primary survey-based study assessing awareness, adoption, and perceived effectiveness of Systematic Investment Plans (SIPs) among retail investors in the Jharkhand region.

Literature Review (9/24): Exploring the potential of AI in personal financial planning and advisory services

2024

- **Relation:** Supports the motivation for our app's AI-driven advisory features (automation, predictive analytics, risk profiling), though it is a conceptual/industry piece rather than a DRL contribution.
- **Methodology:** Discussion-style analysis of AI use cases (automation, predictive analytics, risk assessment) in personal financial planning and advisory workflows.

Literature Review (10/24): Forecasting inflation using disaggregates and machine learning

2023

- **Relation:** Relevant to our goal-cost projection module – our app must inflate future goal costs (e.g. education, car price) accurately, and this paper's ML forecasting approach can inform that sub-module.
- **Methodology:** Aggregates disaggregated (bottom-up) inflation forecasts using linear and nonlinear ML models, applied to Brazilian inflation data, and compares against direct aggregate forecasting.

Literature Review (11/24): From Time Series to Images: A GAF-ResNet-LSTM Framework for Portfolio Optimization

2025

- **Relation:** An alternative deep-learning approach (image-based) to asset selection that could complement our recommendation engine's fund-selection stage, contrasting with the base paper's pure RL approach.
- **Methodology:** Converts price sequences into Gramian Angular Field (GAF) images, extracts spatial features via ResNet and temporal features via LSTM, then applies nested clustered optimization (NCO); validated on NIFTY 50 constituents.

Literature Review (12/24): Hybrid LSTM and PPO Networks for Dynamic Portfolio Optimization

2025

- **Relation:** Directly extends the base paper's PPO approach with a forecasting layer – a natural reference for a possible future enhancement to our recommendation engine.
- **Methodology:** Combines an LSTM forecasting sub-module with a PPO-based allocation agent so decisions are informed by learned temporal price patterns.

Literature Review (13/24): Machine learning in budget forecasting for corporate finance

2024

- **Relation:** While corporate-focused, the proposed ML-driven forecasting architecture is conceptually transferable to our app's personal budget/savings-capacity forecasting module.
- **Methodology:** Conceptual model integrating regression, time-series analysis, and neural networks into a budget forecasting architecture for corporate financial planning.

Literature Review (14/24): Parallel and Multi-Stage Knowledge Graph Retrieval for Financial Asset Recommendations

2025

- **Relation:** Closely relevant to our app's recommendation engine design – shows how to align recommendations with investor behavior, complementing the base paper's reward-based personalization.
- **Methodology:** Knowledge-graph-based retrieval system evaluated on the FAR-Trans dataset, using strict temporal cutoffs to avoid data leakage when generating behaviorally-aligned asset recommendations.

Literature Review (15/24): Regret-Optimized Portfolio Enhancement through DRL and Future Looking Rewards

2025 – Base Paper

- **Relation:** This is the base paper – our project directly adapts its regret-optimized, future-looking reward formulation for goal-based, horizon-sensitive asset allocation.
- **Methodology:** PPO-based DRL framework for dynamic portfolio rebalancing using a regret-based reward (relative to a 60/40 benchmark) combined with a future-looking (forward-horizon) reward component; evaluated via Sharpe ratio and maximum drawdown.

Literature Review (16/24): Risk-Aware Deep Reinforcement Learning for Dynamic Portfolio Optimization

2025

- **Relation:** Closest methodological sibling to the base paper – both use PPO with risk-sensitive rewards, useful for comparing regret-based vs. Sharpe-based reward design in our engine.
- **Methodology:** Integrates a Sharpe-ratio-based reward with direct risk controls (maximum drawdown, volatility constraints), trained using PPO; benchmarked against mean-variance and equal-weighted portfolios.

Literature Review (17/24): Robo Advising and Investor Profiling

2024

- **Relation:** Directly relevant to our risk-profiling module – proposes an objective, quantifiable risk measure that our app's questionnaire-based profiling could adopt instead of vague risk buckets.
- **Methodology:** Introduces an investor profiling method based solely on Relative Risk Aversion (RRA), deriving optimal allocations and comparing them against real robo-advisor (Riskalyze) portfolios for conservative, moderate, and aggressive profiles.

Literature Review (18/24): Robo-advisors in financial services: Redefining wealth management in the age of AI

2026

- **Relation:** Provides a structured typology of robo-advisor institutional forms, useful for positioning our app's design choices within the broader robo-advisory landscape.
- **Methodology:** Conceptual model (grounded in Dubin's theory-building framework) categorizing robo-advisors into Wealth Management Firm, Regulatory, and Personal types, linking institutional design to return maximization, risk minimization, and financial inclusion outcomes.

Literature Review (19/24): Scoring the Ethics of AI Robo-Advice: Why We Need Gateways and Ratings

2025

- **Relation:** Important for our project's ethical/regulatory considerations – highlights trust and transparency issues our app must address when recommending investment allocations.
- **Methodology:** Proposes a regulatory framework requiring robo-advisors to clear four licensing gateways for an AI Robo-Advice License (AIRAL), followed by ongoing ethical-compliance monitoring via a balanced scorecard rating.

Literature Review (20/24): Sentiment-Aware Portfolio Optimization: CVaR-Based Diversification With DRL

2025

- **Relation:** Shares the risk-control philosophy of the base paper (CVaR vs. regret) and shows how sentiment can be incorporated into the state space, a possible future extension for our engine.
- **Methodology:** Uses DDPG to dynamically adjust allocations under Conditional Value-at-Risk (CVaR) constraints while incorporating FinBERT-derived sentiment scores into the state representation, accounting for transaction costs.

Literature Review (21/24): Simul-RL Portfolio Framework: Black-Scholes-Merton and RL for Asset Allocation

2025

- **Relation:** Addresses a limitation relevant to our project – limited historical data for personal goal scenarios – by showing how simulated data can augment RL training, unlike the base paper which relies on historical market data alone.
- **Methodology:** Uses Monte Carlo simulation data (grounded in the Black-Scholes-Merton model) to increase training data complexity and diversity for an RL-based asset allocation agent, addressing overfitting and limited statistical structure in real data.

Literature Review (22/24): The Impact of the Digital Economy on Investment Portfolio Management

2024/25*

- **Relation:** Provides macro-level context on how digital transformation is reshaping portfolio management practices, motivating our project's digital-first (mobile app) approach.
- **Methodology:** Empirical/analytical study examining how digital economy indicators relate to investment portfolio management and investor behavior in stock markets.

Literature Review (23/24): The role of robo-advisors in behavioural finance, shaping investment decisions

2020

- **Relation:** Relevant to our app's UX and risk-communication design – highlights behavioral biases robo-advisors must account for, complementing the base paper's purely quantitative reward design.
- **Methodology:** Longitudinal case study examining robo-advisors through the lens of behavioural finance and their role in shaping investors' financial decision-making processes.

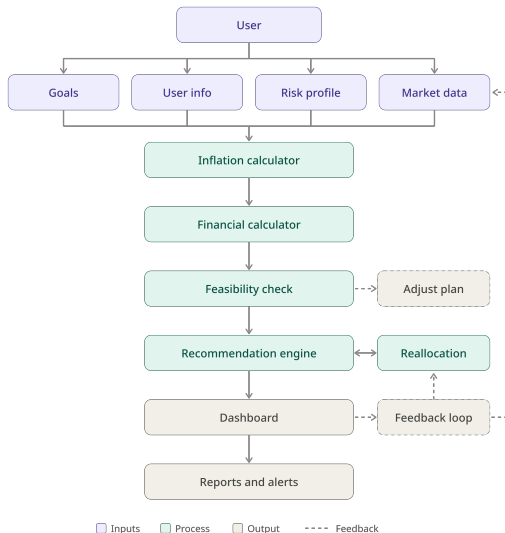
Literature Review (24/24): Uncertainty-Aware Reinforcement Learning for Portfolio Optimization

2024

- **Relation:** Extends the base paper's risk-awareness with explicit uncertainty quantification (Bayesian forecasting), a potential enhancement for our engine's confidence-aware recommendations.
- **Methodology:** Proposes the Uncertainty-Aware Markov Decision Process (UAMDP), coupling Bayesian forecasting, posterior-sampling RL (Thompson sampling), and CVaR-constrained planning; improves Sharpe ratio and roughly halves maximum drawdown vs. baselines.

- **Robo-advisory and AI financial planning literature** covers general AI-driven advisory concepts, investor profiling, ethics, and Indian SIP adoption – but none integrate a *dynamic, reward-shaped* recommendation engine tied to a specific goal's remaining time horizon.
- **No existing system combines:**
 - ① Goal-based horizon-aware planning (amount, deadline, inflation-adjusted target),
 - ② A mobile-first, India-relevant recommendation engine spanning mutual funds, equity, and debt instruments.
- **This project's contribution:** bridging goal-based financial planning with regret-based, future-looking DRL reward shaping in a single mobile recommendation system.

System Architecture



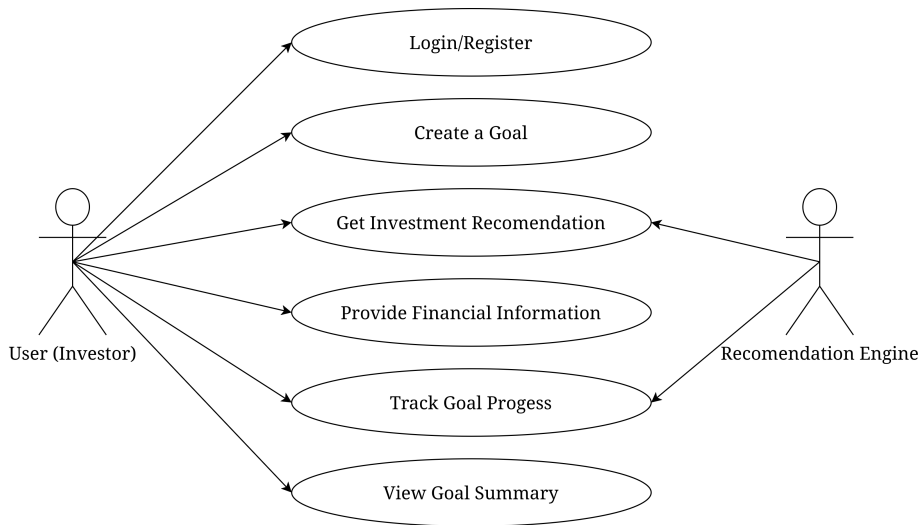
Project Planning: Project Calendar

Phases	July 2026	August 2026	September 2026	October 2026	November 2026	December 2026	January 2026	February 2026
Requirement Analysis and Literature Review								
System Design								
Frontend Development								
Backend Development								
Recommendation Engine Development								
Integrating Modules								
Testing and Bug Fixes								
Deployment and Documentation								

Project Planning: Phase-wise Timeline

- **July 1 – August 10:** Requirement analysis, literature review finalization
- **August 11 – September 5:** System design – UML diagrams, architecture
- **September 6 – November 15:** Frontend Development
- **September 6 – November 15:** Backend development Development
- **September 6 – November 25:** Recommendation engine development & training
- **November 26 – December 25:** Integration of frontend, backend, and recommendation engine
- **December 21 – February 5:** Testing, bug fixing, and evaluation against benchmarks
- **January 26 – February 28:** Documentation and final presentation preparation

Usecase Diagram



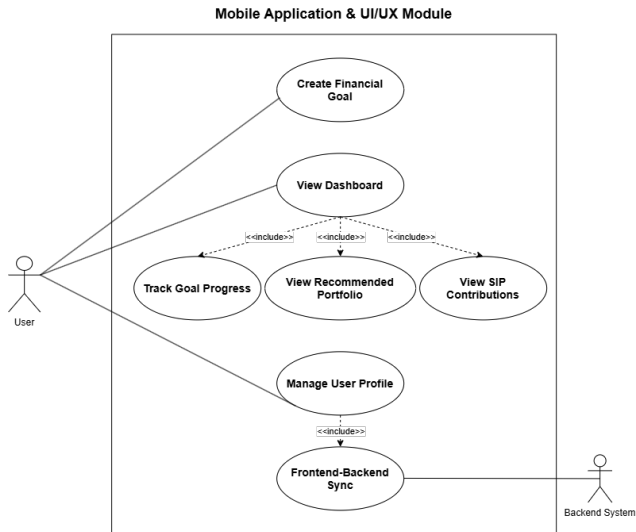
Primary:

- UI/UX architecture
- Mobile screens
- Goal creation interface
- Dashboard
- Recommendation visualization

Supporting:

- Frontend-backend integration
- User testing

Module 1 Use Case Diagram



Risk Profiling & User Modelling

Primary:

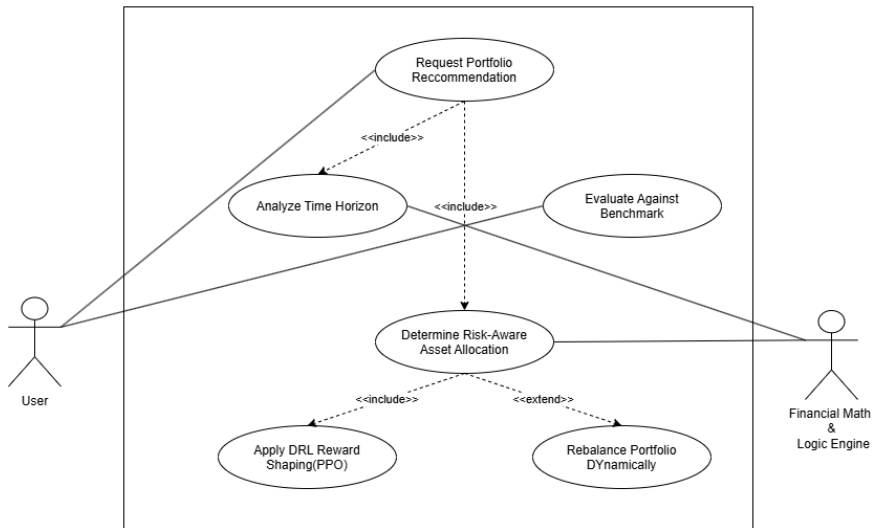
- Risk questionnaire
- Risk tolerance
- Risk capacity
- Risk classification
- User financial profile

Supporting:

- Risk-related literature analysis
- Testing different user profiles

Module 2 Use Case Diagram

AI Recommendation & Portfolio Allocation Module



AI Recommendation & Portfolio Allocation

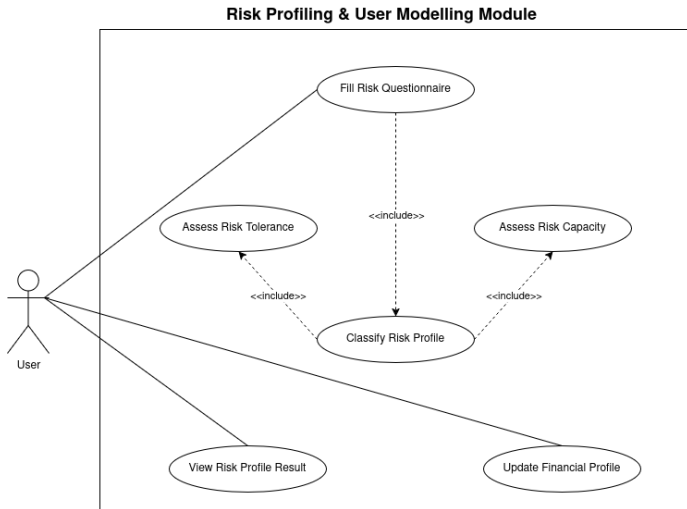
Primary:

- Recommendation engine
- Time-horizon analysis
- Asset allocation
- Dynamic portfolio adjustment
- Base paper's DRL concepts

Supporting:

- PPO/reward mechanism study
- Benchmark comparison
- Recommendation evaluation

Module 3 Use Case Diagram



Financial Forecasting & Financial Computation

Primary:

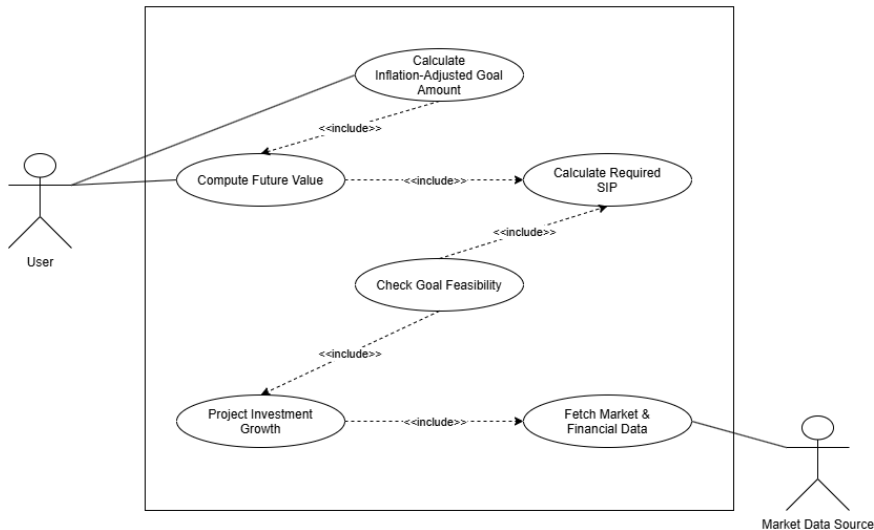
- Inflation-adjusted goal amount
- Future Value calculation
- SIP calculation
- Expected investment growth
- Goal feasibility
- Financial/market data processing

Supporting:

- Forecasting model/data analysis
- Financial API integration
- Validation of financial calculations

Module 4 Use Case Diagram

Financial Forecasting & Computational Module



Modular Breakdown

Module 1: Mobile Application & UI/UX

- UI/UX architecture, Mobile Screens, Recommendation visualization.
- Goal Creation Interface, Dashboard.

Module 2: AI Recommendation and Portfolio Allocation

- Recommendation Engine, Time Horizon Analysis.
- Asset Allocation, Portfolio Reallocation.

Module 3: Risk Profiling and & Modelling

- Risk Questionnaire, Risk Tolerance, Risk Capacity.
- Risk Classification, User financial Profile.

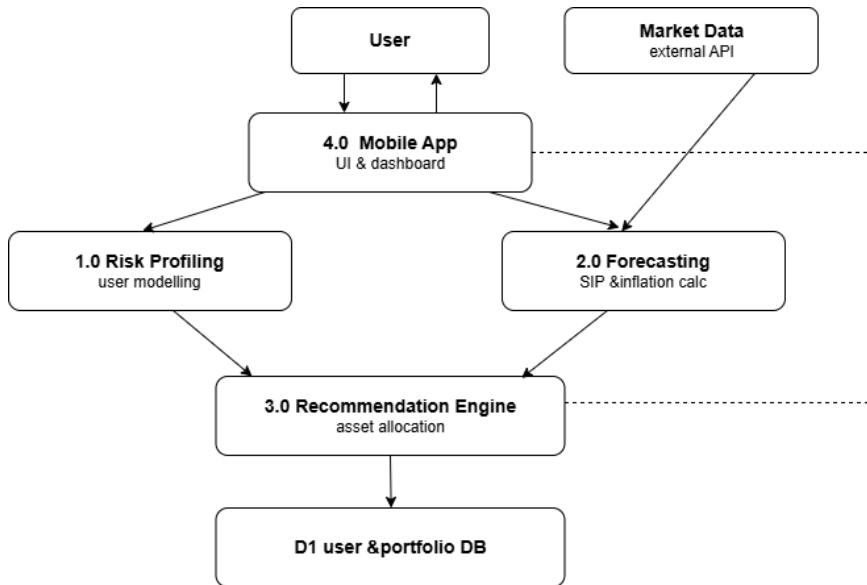
Module 4: Financial Forecasting & Financial Computation

- Inflation adjusted Goal amount, SIP Calculation.
- Expected Investment Growth, Market Data Processing.

Level 0: Dataflow Diagram



Level 1: Dataflow Diagram



Tools and Technologies Used

- **Frontend (Mobile App):** Flutter
- **Backend:** Python (FastAPI)
- **Database:** Supabase (Postgres)
- **Recommendation Engine:** Python, PyTorch / Stable-Baselines3
- **Version Control:** Git & GitHub

Resource Requirements

- **Human Resources:** 4 team members (UI/UX, risk profiling, AI recommendation, financial computation) under faculty guidance.
- **Hardware:** Team laptops (8–16 GB RAM) and personal Android devices for testing; optional free GPU notebooks (Kaggle/Colab) for PPO training.
- **Software (all free/open-source):** Flutter, Python, FastAPI, PyTorch, Stable-Baselines3, Git/GitHub.
- **Cloud Services:** Supabase (Postgres, Auth) and Render for FastAPI hosting – both on free tiers.
- **Data:** Free AMFI-sourced mutual fund NAV data via MFapi.in (no API key required).

Cost Estimation

Item	Plan / Basis	Baseline	Recommended
Database (Supabase)	Free: 500 MB, 50K MAU	0	0
Backend hosting (Render)	Free tier; Starter \$7/mo $\times$ 3 months for demo	0	1,800
RL training compute	Kaggle free; Colab Pro \$9.99/mo $\times$ 3 months	0	2,500
Market data (MFapi.in)	Free, rate-limited (cache)	0	0
Play Console	One-time \$25 (optional; APK for demo)	0	2,100
Development tools	Open-source	0	0
Total (INR)		0	$\approx$ 6,400

Functional Requirements

- **Financial Goal Management:** Allow users to define financial goals with target amount and time horizon.
- **Risk Profiling:** Assess the user's risk tolerance and risk capacity to classify the user's risk profile.
- **Financial Calculation:** Calculate inflation-adjusted goal amount, required SIP contribution, and expected investment growth.
- **Investment Recommendation:** Recommend suitable allocation across mutual funds, equity, and debt instruments based on the user's goal and risk profile.
- **Dynamic Portfolio Adjustment:** Adjust asset allocation based on the remaining goal horizon, risk profile, and changing market conditions.

Non-Functional Requirements

- **Usability:** The application should provide a simple and intuitive mobile-first interface.
- **Performance:** Financial calculations and investment recommendations should be generated efficiently.
- **Security:** User personal and financial information should be securely handled and stored.
- **Reliability:** Financial calculations and recommendation results should be consistent and dependable.
- **Accuracy:** Goal calculations, SIP calculations, inflation adjustments, and portfolio recommendations should provide reliable results.

Conclusion

- Existing goal-based financial planning tools rely on static, rule-based risk assessment that does not adapt to a goal's remaining time horizon or evolving market conditions.
- The reviewed literature demonstrates strong individual advances in robo-advisory design, and behavioral/ethical considerations – but no single system unifies these for **personal, goal-based** investment planning.
- This project proposes a mobile application that adapts the base paper's regret-optimized, future-looking reward mechanism to recommend dynamic investment allocations for individual financial goals.
- This phase 1 review establishes the problem statement, objectives, literature foundation, system design, Modular division and Technological requirements.

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Thank You